E-Commerce Data Analysis Project

Project Documentation

1. Project Overview

This project delivers an end-to-end data analytics workflow for an e-commerce retail business, covering the full pipeline from raw data to actionable business insights. The workflow includes acquiring a real-world, multi-table e-commerce dataset, cleaning and transforming it into an analysis-ready star schema using Power Query, exploring the data with SQL to answer a defined set of business questions, and building an interactive Power BI dashboard to communicate the results.

The dataset is organized around a fact table of individual sales transactions, linked to five supporting dimension tables (customer, item, store, time and transaction/payment). This star-schema structure supports flexible slicing and aggregation of revenue and sales-volume metrics across customers, products, locations, payment methods and time periods.

2. Business Problem / Analytical Objective

The overall objective of this project is to analyze historical e-commerce transaction data to understand how revenue is generated and distributed across the business, and to translate that understanding into practical, data-driven recommendations.

Specifically, the analysis aims to:

- Measure overall business performance in terms of revenue, order volume and pricing.
- Understand how revenue is distributed across customers, and whether the business depends on a small number of high-value customers.
- Track how revenue changes over time (year-over-year and month-over-month) to identify growth trends and seasonality.
- Evaluate product performance to determine which products drive revenue, and whether sales volume and revenue are aligned.
- Provide clear, actionable recommendations for customer retention, product strategy and revenue management.

3. Dataset Description and Source

3.1 Source

The dataset used in this project is the publicly available "ECommerce Data Analysis" dataset published on Kaggle by the user mmohaiminulislam. It models a full e-commerce sales pipeline and is provided as a set of six related CSV files representing a star schema (one fact table and five dimension tables).

Dataset link:

https://www.kaggle.com/datasets/mmohaiminulislam/ecommerce-data-analysis

3.2 Dataset Structure

The dataset consists of one central fact table (fact_table.csv) holding one row per sales transaction, and five dimension tables that describe the customers, items, stores, time periods and payment methods referenced by each transaction:

File	Rows	Role	Description
fact_table.csv	1,000,000	Fact	Transaction-level sales records: payment, customer, time, item and store keys, quantity, unit, unit price and total price.
customer_dim.csv	9,191	Dimension	Customer identifiers with name, contact number and national ID (NID).
item_dim.csv	264	Dimension	Product/item catalog with name, description, unit price, manufacturing country, supplier and unit of sale.
store_dim.csv	726	Dimension	Store location hierarchy: division, district and sub-district.
time_dim.csv	99,999	Dimension	Date/time breakdown per transaction: hour, day, week, month, quarter and year.
trans_dim.csv	39	Dimension	Payment method reference: transaction type (cash, card, mobile) and bank name.

Together, these tables form a star schema: fact_table.csv sits at the center and connects to each dimension table through a shared key (coustomer_key, item_key, store_key, time_key and payment_key), enabling revenue and quantity metrics to be aggregated and filtered by customer, product, location, date/time and payment method.

4. Data Cleaning and Preparation Process

4.1 Objective

The objective of this data-cleaning process was to prepare the six CSV datasets for analysis in Power BI. The datasets were cleaned and transformed using Power Query by handling missing values, checking duplicates, correcting data types, standardizing inconsistent values and formats, and creating the required combined date-time field.

The six datasets processed were:

1. fact_table.csv
2. customer_dim.csv
3. item_dim.csv
4. store_dim.csv
5. time_dim.csv
6. trans_dim.csv

4.2 fact_table.csv

Missing Values

- The unit column contained missing values.
- Missing unit values were replaced with the most frequent unit, ct.
- No other columns contained missing values.

Duplicate Records

- Duplicate records were checked.
- No duplicate records were found.

Data Types

The data types were reviewed and corrected where necessary:

- Key columns were set to Text.
- quantity was set to Whole Number.
- unit was set to Text.
- unit_price was set to Decimal Number.
- total_price was set to Decimal Number.

Inconsistent Values

The unit column contained inconsistent representations. The following standardizations were performed:

- ct. → ct
- oz. → oz
- botlltes → bottles

- Upper/lowercase variations such as Bags and Ct were standardized using lowercase formatting.
- pk → pack to maintain consistency with the corresponding item dimension.

Data Validation

The total_price column was validated against:

quantity × unit_price

No inconsistencies were found.

Result

The fact table was prepared as a consistent, analysis-ready transaction dataset while retaining the original records.

4.3 customer_dim.csv

Missing Values

- The name column contained 27 missing values.
- These values were replaced with Unknown rather than removing the customer records.
- This preserves the customer keys and maintains referential integrity with the fact table.

Duplicate Records

- Duplicate records were checked.
- No duplicate records requiring removal were identified.

Data Types

- coustomer_key → Text
- name → Text
- contact_no → Text
- nid → Text

The contact number and NID were stored as Text because they are identifiers rather than numerical values, and storing them as numbers could cause formatting or leading-zero issues.

Text Cleaning

- Text fields were reviewed and cleaned using Trim and Clean where appropriate.
- Customer keys were preserved as identifiers.

Result

The customer dimension contains complete customer records with missing names handled appropriately and identifiers stored using suitable data types.

4.4 item_dim.csv

Missing Values

- The unit column contained 1 missing value.

- The missing value was replaced with the most frequent unit, ct.

Duplicate Records

- Duplicate records were checked.
- No duplicate records were found.

Data Types

The columns were assigned appropriate types:

- item_key → Text
- item_name → Text
- desc → Text
- unit_price → Decimal Number
- man_country → Text
- supplier → Text
- unit → Text

Inconsistent Values

The unit column contained inconsistent representations. These were standardized:

- ct. → ct
- oz. → oz
- botlltes → bottles
- Bags → bags
- Ct → ct
- pk → pack

The pk and pack values were standardized to pack after reviewing the item descriptions and determining that they represented the same packaging unit.

Country Formatting

- Manufacturer country names were standardized using consistent capitalization.
- For example, lowercase country names such as poland were standardized to Poland.

Result

The item dimension was standardized and prepared for consistent analysis and integration with the fact table.

4.5 store_dim.csv

Header Correction

The original file did not initially use the first row as the column headers. The first row contained the intended field names:

- store_key
- division

- district
- upazila

The first row was promoted to headers in Power Query.

The upazila column was renamed to sub_district to make the field easier to understand for analysis.

Missing Values

- The four columns were checked for missing values.
- No missing values were found.

Data Types

All columns were set to Text:

- store_key
- division
- district
- sub_district

Duplicate Records

- store_key was checked for duplicates.
- Duplicate geographic values were not treated as duplicates because multiple stores can legitimately operate within the same district or sub-district.

Text Cleaning

- Trim was applied to remove unnecessary leading and trailing spaces.
- Clean was applied to remove unwanted hidden/non-printable characters.
- The existing uppercase format of the geographic fields was retained because it was consistent throughout the dataset.

Result

The store dimension was cleaned and prepared as a consistent location reference table.

4.6 time_dim.csv

Date and Time Transformation

The original date-time field contained values such as:

01-01-2015 08:14

Because direct conversion caused date-format errors, the date and time components were temporarily separated.

The date component was converted using Using Locale → English (United Kingdom) to correctly interpret the DD-MM-YYYY format.

The date and time components were then combined into a new date_time column.

Data Types

The final fields were assigned appropriate types:

- time_key → Text
- date_time → Date/Time
- hour → Whole Number
- day → Whole Number
- week → Text
- month → Whole Number
- quarter → Text
- year → Whole Number

Missing Values

All columns were checked for missing values and no missing values were found.

Duplicate Records

The time_key column was checked for duplicate values to ensure that each time record could be uniquely identified.

Validation

The date-time values were checked against the separate date components. For example, a value such as:

01/01/2015 08:14

corresponds to:

- Hour = 8
- Day = 1
- Month = 1
- Quarter = Q1
- Year = 2015

Result

The time dimension was converted into a consistent date-time structure suitable for time-based analysis in Power BI.

4.7 trans_dim.csv

Missing Values

- All three columns were checked.
- No missing values were found.

The transaction types included:

- cash
- card
- mobile

Data Types

- payment_key → Text
- trans_type → Text
- bank_name → Text

The payment key was stored as Text because it is an identifier such as P001, rather than a numerical measurement.

Duplicate Records

- payment_key was checked for duplicates.
- No duplicate payment keys were identified.

Category Consistency

The transaction type categories were reviewed and found to be consistent.

Bank names were retained without manually changing names such as Limited and Ltd, since these represent source values and changing them without an authoritative reference could introduce incorrect information.

Result

The transaction dimension was cleaned and prepared for use as a payment-method reference table.

4.8 Overall Validation

After completing the cleaning process, the six datasets were reviewed in Power Query to ensure that:

- Missing values were appropriately handled.
- Duplicate records were checked.
- Appropriate data types were assigned.
- Inconsistent values were standardized.
- Text formatting was cleaned.
- Date and time values were correctly interpreted.
- Relationships between dimension tables and the fact table were preserved.
- No unnecessary records were removed.
- The datasets were suitable for analysis in Power BI.

Final Outcome

The six CSV files were transformed into clean, consistent, and analysis-ready datasets. The cleaning process preserved valid records while addressing missing values, formatting inconsistencies, data-type issues, and date-time conversion problems.

The cleaned datasets are now ready to be loaded into Power BI for data modeling, visualization, and further analysis.

5. Business Questions

The analysis was conducted to answer the following business questions:

Customer Analysis

- How much does each customer contribute to total revenue?
- Are revenues concentrated among a small number of customers?
- Can customers be segmented into meaningful spending groups?
- Which customer segment generates the highest revenue?

Revenue and Growth Analysis

- Is the business growing over time?
- How does annual revenue change from year to year?
- Are there seasonal patterns in revenue performance?
- Which months consistently perform better or worse?

Product Analysis

- Which products generate the highest revenue?
- Which products sell the largest quantities?
- Does higher sales volume always lead to higher revenue?
- Is revenue concentrated in a small number of products or distributed across the product portfolio?

6. SQL Analysis

The SQL analysis was divided into basic descriptive analysis and advanced business analysis.

6.1 Basic Analysis

The following metrics were calculated:

- Total revenue generated.
- Total quantity sold.
- Average product price.
- Total transactions.
- Number of unique customers.
- Revenue by division and district.
- Revenue by payment method.
- Revenue and transaction counts by year.
- Top-performing customers.
- Top-selling products.

- Peak transaction hours.

SQL concepts used:

- Filtering (WHERE)
- Sorting (ORDER BY)
- Aggregations (SUM, AVG, COUNT, MIN, MAX)
- Grouping (GROUP BY)
- Table relationships (JOIN)

Basic Analysis Queries

```
SELECT item_key, item_name, unit_price
FROM item_dim_cleaned
WHERE unit_price > 10
ORDER BY unit_price DESC;

SELECT SUM(total_price) AS total_revenue
FROM fact_table_cleaned;

SELECT SUM(quantity) AS total_quantity_sold, AVG(unit_price) AS avg_unit_price
FROM fact_table_cleaned;

SELECT COUNT(*) AS total_transactions, COUNT(DISTINCT coustomer_key) AS unique_customers
FROM fact_table_cleaned;

SELECT MIN(unit_price) AS min_price, MAX(unit_price) AS max_price
FROM item_dim_cleaned;

SELECT s.division, SUM(f.total_price) AS total_revenue, SUM(f.quantity) AS total_quantity
FROM fact_table_cleaned f
JOIN store_dim_cleaned s ON f.store_key = s.store_key
GROUP BY s.division
ORDER BY total_revenue DESC;

SELECT s.district, SUM(f.total_price) AS total_revenue
FROM fact_table_cleaned f
JOIN store_dim_cleaned s ON f.store_key = s.store_key
GROUP BY s.district
HAVING SUM(f.total_price) > 1000000
ORDER BY total_revenue DESC;

SELECT f.item_key, i.item_name, SUM(f.quantity) AS total_quantity_sold
FROM fact_table_cleaned f
JOIN item_dim_cleaned i ON f.item_key = i.item_key
GROUP BY f.item_key, i.item_name
ORDER BY total_quantity_sold DESC;

SELECT f.item_key, i.item_name, SUM(f.total_price) AS total_revenue
FROM fact_table_cleaned f
JOIN item_dim_cleaned i ON f.item_key = i.item_key
GROUP BY f.item_key, i.item_name
ORDER BY total_revenue DESC;

SELECT supplier, COUNT(*) AS item_count
FROM item_dim_cleaned
GROUP BY supplier
HAVING COUNT(*) > 5
```

```
ORDER BY item_count DESC;

SELECT man_country, COUNT(*) AS item_count, AVG(unit_price) AS avg_price
FROM item_dim_cleaned
GROUP BY man_country
ORDER BY item_count DESC;

SELECT t.trans_type, COUNT(*) AS transaction_count, SUM(f.total_price) AS total_revenue
FROM fact_table_cleaned f
JOIN trans_dim_cleaned t ON f.payment_key = t.payment_key
GROUP BY t.trans_type
ORDER BY total_revenue DESC;

SELECT t.bank_name, SUM(f.total_price) AS total_revenue, COUNT(*) AS transaction_count
FROM fact_table_cleaned f
JOIN trans_dim_cleaned t ON f.payment_key = t.payment_key
WHERE t.trans_type = 'card' AND t.bank_name <> 'None'
GROUP BY t.bank_name
ORDER BY total_revenue DESC;

SELECT tm.year, SUM(f.total_price) AS total_revenue, COUNT(*) AS transaction_count
FROM fact_table_cleaned f
JOIN time_dim_cleaned tm ON f.time_key = tm.time_key
GROUP BY tm.year
ORDER BY tm.year;

SELECT tm.quarter, SUM(f.total_price) AS total_revenue
FROM fact_table_cleaned f
JOIN time_dim_cleaned tm ON f.time_key = tm.time_key
WHERE tm.year = 2018
GROUP BY tm.quarter
ORDER BY tm.quarter;

SELECT f.coustomer_key, c.name, SUM(f.total_price) AS total_spent, COUNT(*) AS
num_transactions
FROM fact_table_cleaned f
JOIN customer_dim_cleaned c ON f.coustomer_key = c.coustomer_key
GROUP BY f.coustomer_key, c.name
ORDER BY total_spent DESC;

SELECT f.store_key, COUNT(*) AS num_transactions, AVG(f.total_price) AS avg_transaction_value
FROM fact_table_cleaned f
GROUP BY f.store_key
HAVING COUNT(*) > 500
ORDER BY avg_transaction_value DESC;

SELECT tm.hour, COUNT(*) AS transaction_count
FROM fact_table_cleaned f
JOIN time_dim_cleaned tm ON f.time_key = tm.time_key
GROUP BY tm.hour
ORDER BY transaction_count DESC;
```

6.2 Advanced Analysis

Customer Revenue Contribution

Each customer's contribution to total company revenue was calculated.

```
-- ========================================
-- 1- Customer Revenue Contribution
```

```sql
-- How much does each customer contribute
-- to the total revenue?
-- ======================================

SELECT f.coustomer_key,c.name,

    -- Total money spent by the customer
    SUM(f.total_price) AS total_spent,

    -- Percentage of total revenue
    CONCAT(
    ROUND(
        SUM(f.total_price) * 100.0 /
        SUM(SUM(f.total_price)) OVER (),2) , '%') AS revenue_percentage

FROM fact_table_cleaned f

JOIN customer_dim_cleaned c
    ON f.coustomer_key = c.coustomer_key

GROUP BY
    f.coustomer_key,
    c.name

ORDER BY total_spent DESC;


-- Know Range of Total Revenue Percentage
WITH customer_sales AS
(
    SELECT
        f.coustomer_key,
        SUM(f.total_price) AS total_spent
    FROM fact_table_cleaned f
    GROUP BY f.coustomer_key
),
customer_pct AS
(
    SELECT
        coustomer_key,
        total_spent * 100.0 /
        SUM(total_spent) OVER () AS revenue_percentage
    FROM customer_sales
)
SELECT
    MIN(revenue_percentage) AS MinPct,
    MAX(revenue_percentage) AS MaxPct
FROM customer_pct;
```

Key Finding

The highest individual customer contributed approximately 1.62% of total revenue.

Insight

Revenue is broadly distributed across customers, indicating that the business is not dependent on a small number of customers.

Customer Segmentation

Customers were divided into three spending groups using the NTILE() window function:

- Low Value Customers
- Medium Value Customers
- High Value Customers

```
-- ========================================
-- 2- Customer Spending Segmentation
-- Divide customers into 3 groups based on
-- their actual spending distribution
-- ========================================

WITH customer_sales AS
(
    SELECT f.coustomer_key, c.name,

        -- Total spending for each customer
        SUM(f.total_price) AS total_spent

    FROM fact_table_cleaned f

    JOIN customer_dim_cleaned c
        ON f.coustomer_key = c.coustomer_key

    GROUP BY f.coustomer_key, c.name), customer_groups AS
(
    SELECT coustomer_key,name, total_spent,

        -- Divide customers into 3 groups
        -- based on their spending
        NTILE(3) OVER
        (
            ORDER BY total_spent
        ) AS spending_group

    FROM customer_sales
)

SELECT coustomer_key,name,total_spent,

    -- Give each group a meaningful business name
    CASE
        WHEN spending_group = 1
            THEN 'Low Value'

        WHEN spending_group = 2
            THEN 'Medium Value'

        WHEN spending_group = 3
            THEN 'High Value'
    END AS customer_segment

FROM customer_groups

ORDER BY total_spent DESC;

-- ========================================
-- 3-Summary of Customer Segments
-- ========================================
```

```
WITH customer_sales AS
(
    SELECT f.coustomer_key,
        SUM(f.total_price) AS total_spent

    FROM fact_table_cleaned f

    GROUP BY f.coustomer_key),customer_groups AS
(
    SELECT coustomer_key, total_spent,

        NTILE(3) OVER
        (
            ORDER BY total_spent
        ) AS spending_group

    FROM customer_sales
)

SELECT
    CASE
        WHEN spending_group = 1 THEN 'Low Value'
        WHEN spending_group = 2 THEN 'Medium Value'
        WHEN spending_group = 3 THEN 'High Value'
    END AS customer_segment,

    COUNT(*) AS customer_count,
    SUM(total_spent) AS total_revenue

FROM customer_groups

GROUP BY spending_group

ORDER BY spending_group;
```

Segment Summary

Segment	Customers	Revenue
Low Value	3,064	30.6M
Medium Value	3,064	35.0M
High Value	3,063	39.8M

Insight

Although High Value customers represent only one-third of the customer base, they generate the highest share of revenue.

Year-over-Year Revenue Analysis

Annual revenue was compared with the previous year using the LAG() window function.

```
-- ========================================
-- 4- Year-over-Year Revenue Growth
-- Compare each year with the previous year
-- ========================================

WITH yearly_sales AS
```

```
(
    SELECT tm.year,

        -- Total revenue for each year
        SUM(f.total_price) AS total_revenue

    FROM fact_table_cleaned f

    JOIN time_dim_cleaned tm
        ON f.time_key = tm.time_key

    WHERE tm.year < 2021

    GROUP BY tm.year
)

SELECT year,total_revenue,

    -- Revenue of the previous year
      LAG(total_revenue) OVER
    (
        ORDER BY year
    ) AS previous_year_revenue,

    -- Difference between current and previous year
    total_revenue
    - LAG(total_revenue) OVER
    (
        ORDER BY year
    ) AS revenue_change

FROM yearly_sales

ORDER BY year;


-- Orders count for each year --
SELECT year, COUNT(*) AS orders_count
FROM fact_table_cleaned f
JOIN time_dim_cleaned t
    ON f.time_key = t.time_key
WHERE year < 2021

GROUP BY year
ORDER BY year;
```

Key Findings

- Revenue remained relatively stable between 2014 and 2020.
- Annual revenue averaged approximately 15 million.
- The highest revenue was recorded in 2018.
- No consistent upward growth trend was observed.

Insight

The business appears mature and stable rather than rapidly growing.

Monthly Revenue Growth Analysis

Monthly revenue changes were analyzed to identify seasonality patterns.

```
-- ========================================
-- 5 - Monthly Revenue Growth
-- Compare each month with the previous month
-- ========================================

WITH monthly_sales AS
(
    SELECT t.year,t.month,

        -- Total revenue for each month
        SUM(f.total_price) AS monthly_revenue

    FROM fact_table_cleaned f

    JOIN time_dim_cleaned t
        ON f.time_key = t.time_key

    WHERE t.year < 2021

    GROUP BY t.year,t.month
),

monthly_comparison AS
(
    SELECT year,month,monthly_revenue,

        -- Revenue from the previous month
        LAG(monthly_revenue) OVER
        (
            ORDER BY year, month
        ) AS previous_month_revenue

    FROM monthly_sales
)

SELECT year,month,monthly_revenue,previous_month_revenue,

    -- Calculate the change in revenue
    monthly_revenue - previous_month_revenue
        AS revenue_change,

    -- Calculate percentage growth
   CONCAT( ROUND(
        (monthly_revenue - previous_month_revenue)
        * 100.0
        / NULLIF(previous_month_revenue, 0),
        2
    ),'%') AS growth_percentage

FROM monthly_comparison

ORDER BY year,month;

-- Orders count for each month in each year
SELECT t.year,t.month,
    COUNT(*) AS orders_count
FROM fact_table_cleaned f
JOIN time_dim_cleaned t
```

```
    ON f.time_key = t.time_key
WHERE t.year < 2021
GROUP BY t.year,t.month
ORDER BY t.year,t.month;
```

Key Findings

- February consistently showed revenue declines.
- March regularly showed strong recovery.
- May and December often delivered strong performance.

Insight

The business exhibits recurring seasonal patterns despite stable annual performance.

Monthly Order Analysis

Monthly order counts were analyzed to explain revenue fluctuations.

Key Finding

Order volume decreased significantly during February across multiple years.

Insight

The February revenue decline is primarily driven by lower order volume rather than pricing changes.

Product Performance Analysis

Product sales quantity, revenue, and revenue per unit were evaluated.

```
-- ========================================
-- 6- Product Performance
-- Compare sales volume with revenue
-- ========================================

SELECT f.item_key, i.item_name,

    -- Number of units sold
    SUM(f.quantity) AS total_quantity,

    -- Revenue generated
    SUM(f.total_price) AS total_revenue,

    -- Average revenue generated per sold unit
    ROUND(
        SUM(f.total_price) * 1.0 /
        NULLIF(SUM(f.quantity), 0),
        2
    ) AS revenue_per_unit

FROM fact_table_cleaned f

JOIN item_dim_cleaned i
    ON f.item_key = i.item_key

GROUP BY
    f.item_key,
    i.item_name
```

```
ORDER BY total_revenue DESC;
```

Key Findings

- Coffee products, K-Cups, energy drinks, and protein drinks generated the highest revenues.
- Products with similar sales quantities often generated significantly different revenues.
- Product pricing plays a major role in overall revenue generation.

Insight

Higher sales volume does not necessarily result in higher revenue. Product value is an important revenue driver.

Product Revenue Contribution Analysis

Each product's contribution to total revenue was calculated.

```
-- ========================================
-- 7- Product Revenue Contribution
-- Identify whether revenue is concentrated
-- in a few products or distributed across
-- the product portfolio
-- ========================================

SELECT i.item_name,
    SUM(f.total_price) AS total_revenue,

    ROUND(
        SUM(f.total_price) * 100.0 /
        SUM(SUM(f.total_price)) OVER (),
        2
    ) AS revenue_percentage

FROM fact_table_cleaned f
JOIN item_dim_cleaned i
    ON f.item_key = i.item_key

GROUP BY i.item_name

ORDER BY total_revenue DESC;
```

Key Findings

- The highest revenue-generating product contributed only 1.24% of total revenue.
- The top revenue-generating products were primarily coffee and energy-drink products.

Insight

Revenue is highly diversified across the product portfolio, reducing dependency on any single product.

7. KPIs Used in the Dashboard

The cleaned star schema was loaded into Power BI to build an interactive report of 13 dedicated DAX measures, organized in a central _Measures table, and used across three report pages. This section documents the data model, the KPI/DAX measures, and the layout of each report page.

7.1 Data Model

The Power BI data model mirrors the star schema from the cleaning stage: fact_table_cleaned sits at the center, with a one-to-many relationship from each dimension table to the fact table:

- customer_dim_cleaned.coustomer_key → fact_table_cleaned.coustomer_key
- item_dim_cleaned.item_key → fact_table_cleaned.item_key
- store_dim_cleaned.store_key → fact_table_cleaned.store_key
- time_dim_cleaned.time_key → fact_table_cleaned.time_key
- trans_dim_cleaned.payment_key → fact_table_cleaned.payment_key

All KPI calculations are defined once as DAX measures in a dedicated _Measures table, rather than being scattered across visuals, so every card and chart on every page reads from the same single source of truth.

7.2 KPI Cards and DAX Measures

13 measures were built, grouped into core business metrics and time-intelligence metrics used for period-over-period comparisons.

Core Metrics

Total Sales

```
SUM('fact_table_cleaned'[total_price])
```

Total Quantity

```
SUM(fact_table_cleaned[quantity])
```

Total Transactions

```
COUNTROWS(fact_table_cleaned)
```

Total Customers

```
DISTINCTCOUNT(customer_dim_cleaned[coustomer_key])
```

Average Transaction Value

```
AVERAGE(fact_table_cleaned[total_price])
```

Time-Intelligence Metrics

Built on top of Total Sales using standard DAX time-intelligence patterns over the time_dim_cleaned[date_time] column:

MTD Sales

```
TOTALMTD(
    [Total Sales],
```

```
    time_dim_cleaned[date_time]
)
```

Previous Month Sales

```
CALCULATE(
    [Total Sales],
    DATEADD(time_dim_cleaned[date_time], -1, MONTH)
)
```

PMTD Sales

```
TOTALMTD(
    [Total Sales],
    DATEADD(time_dim_cleaned[date_time], -1, MONTH)
)
```

MoM %

```
DIVIDE(
    [Total Sales] - [Previous Month Sales],
    [Previous Month Sales]
)
```

YTD Sales

```
TOTALYTD(
    [Total Sales],
    time_dim_cleaned[date_time]
)
```

Previous Year Sales

```
CALCULATE(
    [Total Sales],
    SAMEPERIODLASTYEAR(time_dim_cleaned[date_time])
)
```

PYTD Sales

```
TOTALYTD(
    [Total Sales],
    SAMEPERIODLASTYEAR(time_dim_cleaned[date_time])
)
```

YoY %

```
DIVIDE(
    [Total Sales] - [Previous Year Sales],
    [Previous Year Sales]
)
```

7.3 Report Pages

The measures feed three connected report pages, each combining KPI cards with charts and a shared set of slicers:

Page 1 — Sales Overview

- KPI cards: Total Sales, Total Transactions, Average Transaction Value, MoM %, YoY %.
- Clustered column chart — Total Sales by Division.
- Pie chart — Total Sales by Payment Type (trans_type).
- Line chart — YTD Sales trend.
- Slicers: Division / District (hierarchy), Year / Month (hierarchy).

Page 2 — Market & Regional Performance

- KPI cards: Total Sales, Total Transactions.
- Map visual (Azure Map) — Total Sales by Division and District.
- Column chart — Total Sales by Division.
- Funnel chart — Total Sales by Manufacturing Country, drilling down to Supplier.
- Slicers: Bank Name / Transaction Type, Division / District (hierarchy), Year / Month (hierarchy).

Page 3 — Customer & Transaction Insights

- KPI cards: Total Customers, Total Transactions, Total Quantity, Average Transaction Value, YoY %.
- Donut chart — Total Customers by Payment Type (trans_type).
- Clustered bar chart — Top Customers by Sales (by customer name).
- Line chart — Total Sales by Customer.
- Slicers: Division / District (hierarchy), Bank Name / Transaction Type, Year / Month (hierarchy), Customer Name.

7.4 Filters and Interactivity

Every page shares the same slicer logic so a viewer can move between pages without losing context:

- Division / District — a hierarchy slicer present on all three pages, letting a viewer drill from region down to district.
- Year / Month — a hierarchy slicer present on all three pages for period-based filtering.
- Bank Name / Transaction Type — available on the Market & Regional Performance and Customer & Transaction Insights pages, to isolate performance by payment method or bank.
- Customer Name — available on the Customer & Transaction Insights page to focus every visual on a single customer.

All visuals cross-filter by default, so clicking any bar, slice, funnel segment or map region filters the rest of the page.

8. Key Findings and Insights

- Revenue remained stable between 2014 and 2020 at approximately 15 million annually.
- Order volume remained relatively stable at around 142,000 orders per year.
- The business shows signs of stability rather than rapid growth.
- February consistently underperforms compared with other months.
- Revenue declines in February correspond closely with declines in order volume.
- High Value customers generate the largest share of revenue.
- Revenue is distributed across a broad customer base.
- Coffee products and energy drinks are the strongest revenue drivers.
- Product price significantly influences revenue generation.
- Revenue is diversified across a large product portfolio, reducing business risk.

9. Data-Driven Recommendations

Insights from the Power BI Dashboard

Beyond the SQL analysis, exploring the data interactively in Power BI surfaced three additional insights that a flat query does not show as clearly:

- Geographic concentration (Map visual): Dhaka alone accounts for ~38.7% of total revenue, and Dhaka + Chittagong together contribute ~57.5% — visibly forming a single geographic hotspot on the map rather than an even regional spread.
- Single-supplier sourcing per country (Country → Supplier funnel): drilling from any manufacturing country down to supplier lands on exactly one supplier every time — 100% of each country's revenue flows through a single supplier, a sourcing concentration that a flat supplier ranking does not reveal.
- Broad payment-channel reach vs. revenue concentration (Donut chart): 100% of customers have used card, 99.97% have used mobile, and 94.0% have used cash — so nearly every customer is reachable through any of the three channels — yet card alone still drives ~89.7% of revenue, mobile ~7.7%, and cash ~2.6%.

Customer Recommendations

- Develop loyalty programs targeting High Value customers.
- Implement personalized promotions for top-spending customers.
- Monitor High Value customers for retention opportunities.

Product Recommendations

- Prioritize inventory availability for top-performing coffee and energy-drink products.
- Increase marketing focus on products with high revenue contribution.
- Expand cross-selling opportunities around high-revenue product categories.

Revenue Recommendations

- Investigate the recurring February decline and prepare seasonal campaigns to reduce its impact.
- Leverage strong-performing months such as March, May, and December through promotional campaigns.
- Monitor monthly revenue trends continuously to identify emerging seasonal changes.
- Keep investing in card-payment reliability and acceptance given its ~89.7% revenue share, while maintaining mobile and cash options that keep the business accessible to virtually all customers.

Business Recommendations

- Maintain the diversified product strategy because revenue is not dependent on a single product.
- Continue broad customer acquisition efforts because revenue is distributed across a large customer base.
- Focus on increasing customer lifetime value rather than relying solely on customer acquisition.

- Qualify a second supplier in key sourcing countries to reduce the single-supplier concentration risk seen in the Country → Supplier funnel.
- Evaluate targeted expansion or marketing investment beyond Dhaka and Chittagong to reduce reliance on the two dominant regions shown on the map.

10. Limitations and Assumptions

Limitations

- Data for 2021 was excluded from trend analysis because the number of recorded orders was substantially lower than previous years, indicating incomplete data.
- The dataset does not include profit, cost, or margin information; therefore, profitability analysis could not be performed.
- Customer demographic information is limited, preventing deeper customer profiling.
- Product category information is not explicitly available, limiting category-level analysis.
- External factors such as promotions, holidays, economic conditions, and competitor activity are not included in the dataset.

Power BI Dashboard Limitations

- The Map visual (Azure Map) plots Division and District names through automatic geocoding; sub-district-level placement was not manually verified, so exact pin positions on the map should be read as approximate, not survey-accurate.
- The Country → Supplier funnel reflects a single snapshot of the dataset — one supplier per manufacturing country — which shows sourcing concentration in this data but does not confirm contractual exclusivity with that supplier.
- All KPI cards and charts are interactive: the figures documented in this report reflect the default, unfiltered view of each page. Applying any slicer (Division, District, Year, Month, Bank Name, Transaction Type, or Customer Name) will change every value on that page accordingly.

Assumptions

- All cleaned records accurately represent actual business transactions.
- The relationships between fact and dimension tables are valid.
- Revenue calculations in the fact table are correct and were validated against: Quantity × Unit Price.
- Missing values handled during the cleaning process do not materially affect overall analytical conclusions.
- The cleaned datasets are representative of normal business operations during the analyzed period.